Lesson 1, Beginner - Intro to Tactile Graphics

Lesson 1, Beginner - Intro to Tactile Graphics

Download this activity

This adventure introduces tactile graphics and art, emphasizing hands-on learning and creativity while introducing the idea of using technology to create tactile graphics. It aims to develop motor skills, spatial awareness, and appreciation for accessible art.

Setup - Activity 1

This activity can be broken down into several elements for lesson planning.

Objectives

- Learn how to create tactile graphics using a Draftsman Tactile Drawing Board.
- Develop foundational motor skills required for holding paper and stylus.
- Recognize and understand basic shapes and line types.
- Begin to develop spatial skills and tactile exploration skills through hands-on activities.

Materials Needed

- Draftsman APH Tactile Drawing Board or APH TactileDoodle
- APH Tactile Drawing Film
- Tactile graphics examples (provided by the teacher)
- Paper
- Stylus or Ballpoint Pen
- Hand sanitizer or wipes to clean up ink/pencil

Alternative Materials

- Sensational Blackboard
- Copy Paper
- Crayon on window screen
- Stencils

- Flip-Over Concept Books: Line Paths
- Lots of Dots: Coloring the Garden

Ideas for Carrying Out This Activity

- Push-into planned art activities
- Collaborate with the art teacher
- Ask parents to help facilitate activities at home to allow the student access to further exploration and practice

Adventure Map: Activity 1

Teaching tip: Provide sufficient time for the student to explore, develop skills, and have fun at each step! Encouraging creativity and personal preferences for drawing as much as possible. Some students may be able to accomplish each step in one session; most students will need several sessions to complete the adventure.

1. Introduction

- Ask your student: What is your favorite graphic? Use guiding questions to get your student to think about graphics, including aspects they may not have thought about before.
  - Why is it your favorite? What is your favorite part about it?
  - How do you think it was created? What would you use to recreate it? How would you make it better?
- Discuss the importance of graphics and how graphics can be used to communicate information.
- Show the students a Draftsman Tactile Drawing Board and briefly explain its purpose.
- Share examples of tactile graphics (e.g., simple shapes, or objects) and discuss how they can be helpful.
- Mention that today, they will have the opportunity to create their own tactile graphics

2. Exploring Tactile Graphics

- Provide your student with a tactile graphic (e.g., a simple shape or object) and let them explore it by touch. Ask them to describe what they feel and what they think the graphic represents.
- Ask the student if there is a picture of something they have wanted to explore as a tactile graphic.
- Offer a 3D manipulative and the related tactile graphic to explore the different representations in 3D and 2D

3. Creating Tactile Graphics - Replication

- Introduce a variety sketching techniques:
  - Drawing Lines — 3:44 duration
  - Drawing Squares — 3:53 duration
  - Drawing Circles — 3:59 duration
- Demonstrate how to use the Draftsman Tactile Drawing Board to create basic shapes (e.g., square, circle, triangle, rectangle).
- Have the students choose one of the tactile graphics examples they explored earlier and try to replicate it on their Draftsman boards.
- Provide guidance and support as needed, emphasizing the importance of pen pressure on the paper, staying in contact with the drawing and suggest pen movement/ drawing techniques depicted in the drawing example videos (contour/ straight ahead drawing vs. sketched lines using small movements).
- Allow students to use available tools to assist as needed (e.g., cut out shapes to trace, stencils, ruler).

4. Creating Tactile Graphics - Original Artwork

- Now, introduce the concept of creating their own original tactile graphics.
- Encourage students to think of a simple object or scene they'd like to draw (e.g., a flower, a house, a smiley face).
- Help the student identify the geometric shapes that could be used to create their object or scene.
- Provide them with the materials and guide them through the process of creating their own tactile graphic.

5. Sharing and Discussion

- Have each student share their replicated and original tactile graphics.
- Encourage them to describe what they've created and explain their thought process.
- Discuss how technology can assist in creating tactile graphics. How can technology help in recreating your drawing? What about if you wanted to change something about your drawing?

6. Conclusion

- Summarize the key points of the adventure.
- Emphasize the importance of tactile graphics and art and how technology can enhance the experience.
- Encourage students to continue exploring tactile graphics both in and out of the classroom, and especially during recreational and leisure activities.

Download the whole thing · **Next:** Activity 2: Descriptive Language and Drawing · **Back to:** Adventure Map

Activity 2, Beginner - Descriptive Language and Drawing

Activity 2, Beginner - Descriptive Language and Drawing

Download this activity

In this adventure, students will explore the intersection of descriptive language and tactile graphics. Students will explore how being specific and precise in creating directions is important in the creation of tactile graphics.

Setup - Activity 2

This activity can be broken down into several elements for lesson planning.

Objectives

- Learn the importance of using descriptive language when creating a set of instructions for creating images.
- Learn the importance of effective communication.
- Create an image and precise directions for someone else to follow to recreate the image.

Materials Needed

- Draftsman APH Tactile Drawing Board or APH TactileDoodle
- APH Tactile Drawing Film
- Stylus or Pen
- Tactile rulers
- Cardstock or Cardboard/ for a straight edge that can be folded

Alternative Materials

- Sensational Blackboard
- Copy Paper

Ideas for Carrying Out This Activity

- Allow students to work with partners if possible.

Adventure Map: Activity 2

Teaching tip: Provide sufficient time for the student to explore, develop skills, and have fun at each step! Encouraging creativity and personal preferences for drawing as much as possible. Some students may be able to accomplish each step in one session; most students will need several sessions to complete the adventure.

1. Introduction

- Ask your student: How does a computer know how to create an image? The answer is "You!" The goal of this question is to get the student to understand that in order to create images, computers just follow directions they are given.
- In the following adventure your student will practice drawing, creating directions to recreate their drawings and try to create drawings when given directions.
- Introduction video — 17:48 duration

2. Create Your Graphic

- Provide your student with drawing materials.
- Ask them to create their own drawings using only five geometric shapes (e.g., square, circle, triangle, rectangle)
- If the student is having difficulty coming up with their own idea, have them start out by trying to draw a house using 4 rectangles and a triangle, like in the introduction video and then have them move on to their own drawing.

3. Explain Your Process

- Write or have your student write what they are doing in specific terms (exact measurement, tactile subjective measurement/ finger width, fractional positioning, clock directions or cardinal directions). Use a list format to organize your step-by-step directions.
- Explain that another student will use the directions to recreate the drawing. The other student should not look at the drawing, only the directions.

4. Switch Roles

- This part works best with a peer, but can be done with the teacher.
- After the students have finished their drawing and instructions, have them give just their instructions to a peer to read and try and replicate the drawing.

- After the students have finished they should compare their drawing to the original.
- If it isn't feasible to complete this activity with a peer. The teacher can try and draw the image based only on the student's directions. Make sure to follow the instructions given, highlighting where the student could be more descriptive.

5. Sharing and Discussion

- Have each student share their replicated and original drawings.
- Encourage them to describe what they've created and explain their process.
  - How did you create the instructions for recreating your drawing? Would you change your process for next time?
  - Was your partner's drawing close to yours? If not, what happened? What could you do differently to fix this?
- Discuss how technology can assist in creating tactile graphics.
  - What benefits does using technology allow? (e.g., copying, deleting, rotating, sharing).

6. Conclusion

- Which role were you better at—listening or describing? Explain why.
- Emphasize the importance of communicating effectively.
- How can tactile graphics play a role in being an effective communicator?

Previous: Activity 1: Intro to Tactile Graphics · Download the whole thing · **Next:** Activity 3: Tactile Colors · **Back to:** Adventure Map

Activity 3, Beginner - Tactile Colors

Activity 3, Beginner - Tactile Colors

Download this activity

In this adventure, students will create pixel art using a grid, braille and textured paper. They will learn about color and using a key while developing tactile discrimination skills.

Setup - Activity 3

This activity can be broken down into several elements for lesson planning.

Objectives

- Learn about color and contrast through tactile exploration.
- Practice tactile discrimination by feeling different textures.
- Use a tactile key to understand how textures represent colors.
- Create pixel art through tactile exploration.

Materials Needed

- Graph paper or grid
- Pixel Paper
- Braille labels
- APH Assorted textured paper (e.g., sandpaper, construction paper, tissue paper)
- Color-By-Texture Marking Mats
- Other textured materials (e.g., screen, fabrics)
- Scissors
- Glue stick

Adventure Map: Activity 3

Teaching tip: Provide sufficient time for the student to explore, develop skills, and have fun at each step! Encouraging creativity and personal preferences for drawing as much as possible. Some students may be able to accomplish each step in one session; most students will need several sessions to complete the adventure.

1. **Introduction**
   - Ask your student: What is your favorite color and why? What do you think that color feels like?
     - Talk about language used to describe texture (e.g., smooth, slimy, rough, bumpy, silky, soft, hard, cold, warm)
     - Talk about language used to describe color (e.g., dark, bright, differences in colors).
   - Show examples of pixel art, discuss what makes it unique.
   - Explain that today they will make pixel art using textures to represent colors

2. **Explore Texture and Contrast**
   - Provide different textured paper samples
   - Have students feel and describe textures
   - Provide different textured surfaces and have students draw on paper placed on top of each surface
   - Discuss contrast between textures

3. **Create a Tactile Key**
   - Introduce the concept of a color wheel or palette
   - Select 3-5 textures to represent colors
   - Label textures with color names
   - Create a tactile key for reference

4. **Plan and Create Pixel Art**
   - Student draws design on graph paper
   - Cut textures into pixel/grid sizes
   - Glue pixels onto grid

5. **Sharing and Discussion**
   - Students share artwork and describe textures
   - Compare student pieces and observe contrasts
   - Discuss experience creating tactile pixel art

6. **Conclusion**
   - Summarize the key points of the adventure.
   - Summary of color, texture, contrast
   - Encourage continued exploration of tactile art

Previous: Activity 2: Descriptive Language and Drawing · Download the whole thing · **Next:** Activity 4: Moving from 3D to 2D · **Back to:** Adventure Map

Moving from 3D to 2D: Basic Shapes

Download this activity

In this activity, students will learn how 3D objects are represented as 2D tactile graphics. The translation from 3D to 2D representations benefits from direct, explicit instruction to help students develop a strong foundation for interpreting tactile graphics.

Setup: 3D to 2D Shapes

This activity can be broken down into several elements for lesson planning.

Objectives:

- Practice exploring information in 3 dimensions: height, width, and depth
- Practice exploring information in 2 dimensions: height and width
- Match features from a 3D object to the corresponding 2D tactile graphic
- Explore how shapes can be used to create a recognizable object

Materials Needed:

- Three handheld-size models of the following shapes, approximately 2 inches in size:
  - A cube
  - A ball (sphere)
  - A 3-sided pyramid
- Three shapes, approximately 2 inches in size: A square, circle, and triangle. Choose your adventure! A teacher can prep these shapes OR this can be an activity for students to create their own shapes:
  - Using a stencil, create and cut out a square, circle, and triangle on construction paper or card stock
  - Use a stencil, and a Sensational Blackboard with copy paper. See Ann Cunningham's video *Creating Pop Out Shapes* (1:55)
  - Use the APH Geometric Drawing Stencils
- APH Draftsman or TactileDoodle
  - APH Tactile Drawing Film

- Stylus or ball point pen
- Glue stick
  - ***Teaching tip:*** Put a piece of tap or a bump dot on the glue stick to keep it from rolling off the table
- Construction paper
- Scissors

Alternate Materials:

- Any tactile drawing board, such as the Sensational Blackboard
- Plain copy paper

Ideas for carrying out this activity:

- Art or math class
- Architecture or shop class
- Extension activity when reading a book about shapes

Adventure Map: 3D to 2D Shapes

Teaching tips: Provide sufficient time for students to explore all materials before beginning the activity. Set up a work space with containers or trays to keep stylus/pens, glue stick, scissors, and drawing materials organized. It is best to use 3D objects and materials that can be sanitized or consumable. Keep hand sanitizer handy and invite students to sanitize hands before beginning the activity!

1. Introduction

- Ask your student about their experiences with the geometric shapes square, circle, triangle
  - What do the shapes remind them of? Where have they seen these shapes in day-to-day life?
- Allow time for your student to explore the various 3D models of each shape (cube, ball, pyramid)
- Discuss the features of each shape. Use descriptive language to describe the number of sides and describe if there are corners or edges to a shape. If your student needs more practice using descriptive language, review

TADA! Activity 2

- Experiment with how a stencil can be used to draw each shape on a tactile drawing board. Support troubleshooting of how to coordinate holding the stencil in place while using enough force to make tangible marks with a pen or stylus. If a student needs more practice using a tactile drawing board, review TADA! Activity 1

2. Working with shapes

- Name each geometric shape: square, circle, triangle

- Ask your student to draw a square, circle, triangle. It is important that students try each of the following methods of drawing shapes. Students can build confidence drawing shapes in all 3 ways:
  - Use a stencil

  - Trace a shape

  - Free draw (remind students that it is OK if their drawings are not perfect like the stencil - drawings do not have to be perfect to be perfectly useful!)

- Ask your student to explore what they have drawn; then ask which 2D symbol they would use to represent the sphere? The cube? The pyramid?

- Create a cutout where the space matches or is slightly larger than the size of the 3D object. Have the student push the object through the cutout to see how the negative space matches the 3D object

- Discuss: When a 3D object is depicted as a 2D picture, it has to be distorted because we have eliminated depth. But, a 3D object can still be sufficiently represented by a 2D symbol/graphic

3. The importance of outlines

- An outline is important when drawing a recognizable shape, but it doesn't have to be perfect to be useful.
  - For example: a hand is a very complex shape, yet you can break it down into geometric shapes and create a recognizable object

- Using a tactile drawing board, have your student draw an outline of their hand

- Next, have your student explore their own hand. You can say:
  - Starting with your palm, which shape would you use to represent the

palm?
    * Using the construction paper, cut a shape out to represent the palm. Select a new piece of paper and glue it on while keeping enough space to glue your thumb on later. Just use a little bit of glue for now to tack it down on the paper - you can move it around later if needed

- Now point to your finger. Which shape would you use to represent the finger?
    * Cut out a shape for each finger. Where does each finger attach to the palm? (All 4 fingers need to be on the same side of palm)

- What's the difference between the shapes of different fingers?

- Now let's look at the thumb - it is not a basic shape!
    * When you are ready to attach it to the hand, look at where it attaches to the palm

- Wow! You have made a really complex shape! Yay!

- ***Teaching tip:*** Introduce the correct vocabulary to identify the different shapes (rectangle, oval, circle, etc.) AND the parts of the hand (palm, fingers, thumb)

4. **Sharing and Discussion**

- Have each student share and appreciate each other's **hand**iwork
    - Encourage students to identify the shapes that they discover on one another's newly constructed hands

    - Identify similarities and differences between one another's hands

- Encourage the students to discuss any surprises or troubleshooting they encountered

- Guide discussion on how construction of their hand did not have to be perfect in order for it to be useful in representing a hand

5. **Conclusion**

- Summarize the shapes that were explored and which 2D symbols were used to represent which 3D objects
    - Circle = ball or sphere

    - Square = cube

  - Triangle = 3-sided pyramid

- Brainstorm
  - What other shapes might be in their immediate environment? These shapes could be used to create other drawings. Have students share these shapes with one another and encourage students to create/share shapes for shared exploration. *Note:* Some students might have gaps in their experiential learning and need hands-on exploration of familiar and unknown shapes
    - Rectangles
    - Stars
    - Teardrop
  - How might shapes be used to recreate other familiar objects?
  - Frequently, shapes are used as icons or symbols. Examples:
    - A Nike swoosh
    - A religious cross
    - Star of David
    - A sun
    - 5-pointed star (can be a star in the sky or a starfish or one state on our flag)

Extension Activities:

- Draw a leaf or lots of different types of leaves
- Draw a favorite toy
- Draw a person from an articulated model (see photo below). Tracing is very difficult because the model moves too easily; instead, tape it down OR analyze what the shape of each segment is and draw the segments in proper orientation or cut them out and reassemble them into a proper relationship to each other.

Previous: Activity 3: Tactile Colors · Download the whole thing · **Next:** Activity 5: Test Activity · **Back to:** Adventure Map

An articulated model is created from cardboard shapes that resemble each limb (forearm, upper arm, neck, torso, head, hips, thighs, and lower leg with foot. Each limb is connected at the joint with a brass paper fastener

Figure 1: An articulated model is created from cardboard shapes that resemble each limb (forearm, upper arm, neck, torso, head, hips, thighs, and lower leg with foot. Each limb is connected at the joint with a brass paper fastener

Activity 5, Beginner - Test Activity

Previous: Activity 4: Moving from 3D to 2D · Download the whole thing ·
Back to: Adventure Map

Previous: Activity 4: Moving from 3D to 2D · Download the whole thing ·
Back to: Adventure Map